

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE****End Semester Examination – Summer 2019****Course: B. Tech in Civil Engineering****Subject Name: Hydraulics - I****Max Marks:60****Date: 30-05-2019****Sem: III****Subject Code: BTCVC303****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

**Q. 1 Solve Any Two of the following.****12**

- A) Explain Newton's law of viscosity. Explain the importance of viscosity in fluid motion. (CO2)  
Convert 1 kg/s-m dynamic viscosity in poise.
- B) The dynamic viscosity of oil, used for lubrication between a shaft and sleeve is 6 poise, the shaft is of diameter 0.4 m and rotates at 190 r.p.m. Calculate the power lost in the bearing for a sleeve length of 90mm. The thickness of oil film is 1.5mm. (CO2)
- C) The surface tension of water in contact with air at 20°C is 0.0725N/m. the pressure inside a droplet of water is to be 0.02N/cm<sup>2</sup> greater than outside pressure, calculate the diameter of droplet of water. (CO2)

**Q.2 Solve Any Two of the following.****12**

- A) Derive an expression for the force exerted on a submerged vertical plane surface by the static liquid and locate the position of centre of pressure. (CO2)
- B) Define the following terms:  
a. Buoyancy (CO2)  
b. Centre of Buoyancy  
c. Meta –centre
- C) A rectangular plane surface 2m wide and 3m deep lies in water in such way that its plane makes an angle of 30° with the free surface of water. Determine the total pressure and position of centre of pressure when the upper edge is 1.5m below the free water surface. (CO2)

**Q. 3 Solve Any One of the following.****12**

- A) 1. Define the terms : a) Velocity potential function, b) Stream function  
2. Define equation of continuity. Obtain an expression for continuity equation for three- dimensional flow. (CO1)

- B)**
1. The diameter of pipe at the sections 1 and 2 are 10 cm and 15 cm respectively. Find the discharge through the pipe if the velocity of water flowing through the pipe at section 1 is 5m/s. Determine also the velocity at section 2. (CO1)
  2. Define the following terms:
    - a. Laminar flow
    - b. Turbulent flow
    - c. Steady flow

**Q.4 Solve Any Two of the following.**

12

- A)** Determine the dimensions of following terms
- a. Kinematic viscosity (CO4)
  - b. Discharge
  - c. force
- B)** What is meant by geometric, kinematic and dynamic similarities? (CO4)
- C)** The pressure drop in an aeroplane model of size 1/10 of its prototype is 80 N/cm<sup>2</sup>. The model is tested in water. Find the corresponding pressure drop in prototype. Take density of air = 1.24 kg/m<sup>3</sup>. The viscosity of water is 0.01 poise while the viscosity of air is 0.00018 poise. (CO4)

**Q. 5 Solve Any One of the following.**

12

- A)** Derive an expression for the velocity distribution for laminar flow through a circular pipe. Also sketch the velocity distribution and shear stress distribution across a section of the pipe. (CO4)
- B)** A crude oil of viscosity 0.97 poise and relative density 0.9 poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 100mm and of length 10m. Calculate the difference of pressure at the two ends of the pipe, if 100kg of the oil is collected in a tank in 30 seconds. (CO4)

**Q.6 Solve Any two of the following.**

12

- A)** The difference in water surface level in two tanks, which are connected by three pipes in series of length 300m, 170m and 210m and of diameters 300mm, 200m and 400mm respectively, is 12m. Determine the rate of flow of water if coefficient of friction are 0.005, 0.0052 and 0.0048 respectively, considering a) minor losses (CO3)
- B)** In above question (Q.6 A) Determine the rate of flow of water by neglecting minor losses. (CO3)
- C)** Define the following terms: (CO3)
- a) Reynold's number, b) Froude's number, c) mach's number.

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